ENDOCRINOLOGY

Male Hypogonadism

Internal Medicine Lecture Series

Pathophysiology

- In the male, the testis serves two principal functions: synthesis of testosterone by the interstitial Leydig cells under the control of luteinising hormone (LH), and spermatogenesis by Sertoli cells under the control of follicle-stimulating hormone (FSH).
- Testosterone levels are higher in the morning and therefore, if testosterone is marginally low, sampling should be repeated with the patient fasted at 0900 hrs.
- There is no equivalent of the menopause in men, although testosterone concentrations decline slowly with age .

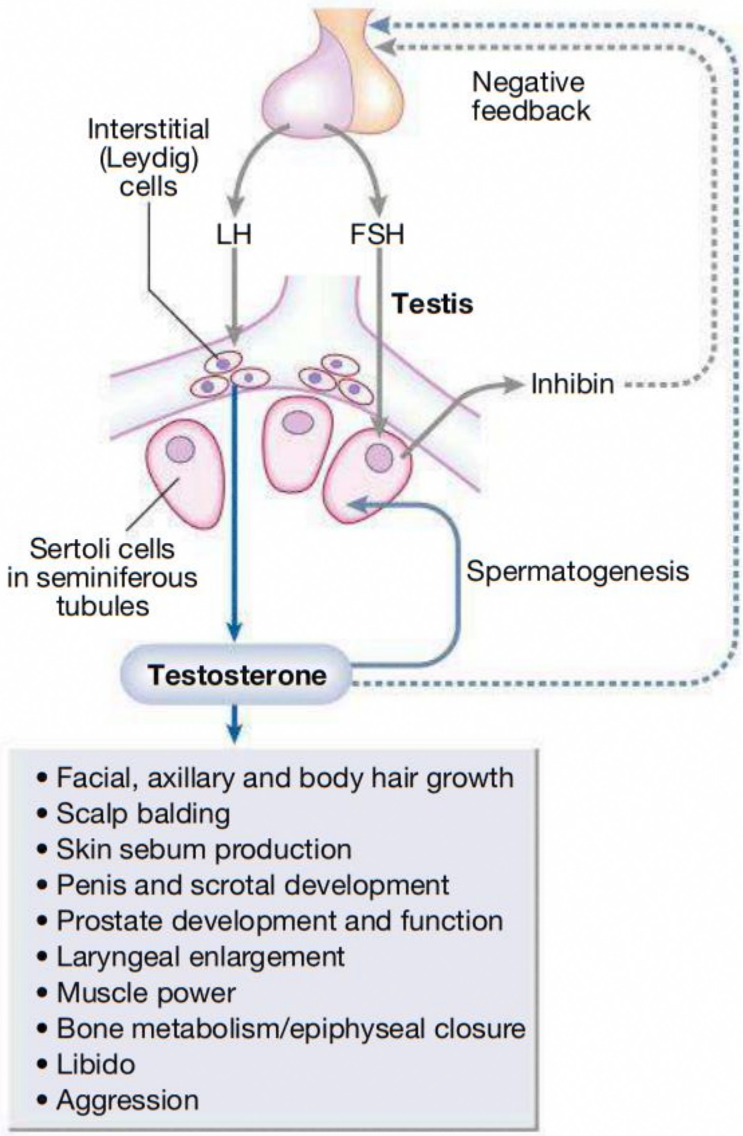


Fig. 20.14 ***Male reproductive physiology. (FSH = follicle-stimulating hormone; L***

1.Hypogonadotrophic hypogonadism:

- This may be due to structural, inflammatory or infiltrative disorders of the pituitary and/or hypothalamus .
- In such circumstances, other pituitary hormones, such as growth hormone, are also likely to be deficient.

Here FSH & LH will be LOW .

Causes :

Hypogonadotrophic hypogonadism

- Structural hypothalamic/pituitary disease (see Box 20.53)
- Functional gonadotrophin deficiency:
  - Chronic systemic illness (e.g. asthma, malabsorption, coeliac disease, cystic fibrosis, renal failure)
  - Psychological stress
  - Anorexia nervosa
  - Excessive physical exercise
  - Hyperprolactinaemia
  - Other endocrine disease (e.g. Cushing's syndrome, primary hypothyroidism)
- Isolated gonadotrophin deficiency (Kallmann syndrome)

2. Hypergonadotrophic hypogonadism

Here FSH & LH will be high

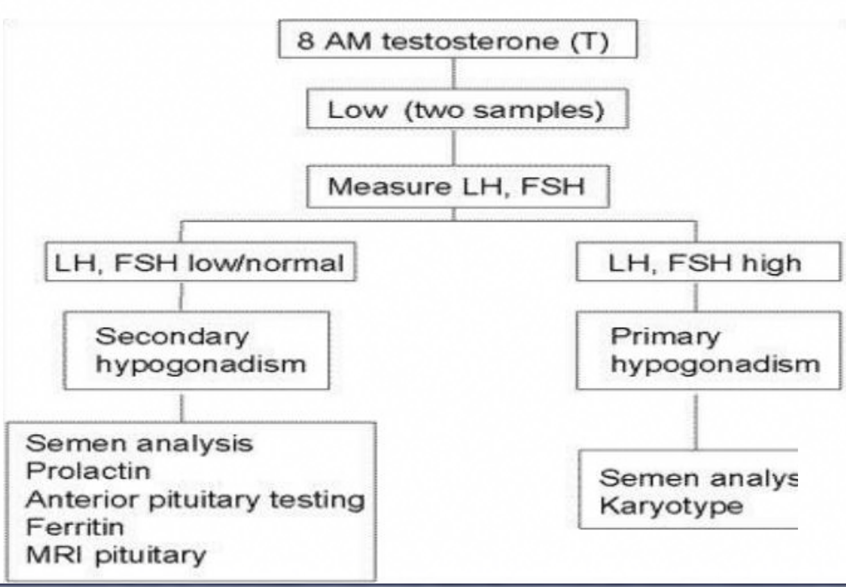
Hypergonadotrophic hypogonadism

- Acquired gonadal damage:
  - Chemotherapy/radiotherapy to gonads
  - Trauma/surgery to gonads
  - Autoimmune gonadal failure
  - Mumps orchitis
  - Tuberculosis
  - Haemochromatosis
- Developmental/congenital gonadal disorders:
  - Steroid biosynthetic defects
  - Anorchidism/cryptorchidism in males
  - Klinefelter syndrome (47,XXY, male phenotype)
  - Turner syndrome (45,X, female phenotype)

- The clinical features of both hypo- and hypergonadotrophic hypogonadism include loss of libido, lethargy with muscle weakness and decreased frequency of shaving.
- Patients may also present with gynaecomastia, infertility, delayed puberty, osteoporosis or anaemia of chronic disease.
- Mild hypogonadism may also occur in older men, particularly in the context of central adiposity and the metabolic syndrome.
- Postulated mechanisms are complex and include reduction in sex hormone-binding globulin by insulin resistance and reduction in GnRH and gonadotrophin secretion by cytokines or oestrogen released by adipose tissue.

Investigations

- Male hypogonadism is confirmed by demonstrating a low fasting 0900-hr serum testosterone level.
- The distinction between hypoand hypergonadotrophic hypogonadism is by measurement of random LH and FSH.
- Patients with hypogonadotrophic hypogonadism should be investigated as described for pituitary disease .
- Patients with hypergonadotrophic hypogonadism should have the testes examined for cryptorchidism or atrophy, and a karyotype should be performed (to identify Klinefelter syndrome).



- Testosterone replacement is clearly indicated in younger men with significant hypogonadism to prevent osteoporosis and to restore muscle power and libido.
- Debate exists as to whether replacement therapy is of benefit in mild hypogonadism associated with ageing and central adiposity, particularly in the absence of structural pituitary/hypothalamic disease or other pituitary hormone deficiency .
- In such instances, a therapeutic trial of testosterone therapy may be considered if symptoms are present (e.g. low libido and erectile dysfunction) and the serum testosterone is consistently low, but the benefits of therapy must be carefully weighed against the potential for harm.

Routes of administration of TST :

- First-pass hepatic metabolism of testosterone is highly efficient, so bioavailability of ingested preparations is poor.

Side effects of TST :

A. Testosterone therapy can aggravate prostatic carcinoma; prostatespecific antigen (PSA) should be measured before commencing testosterone therapy in men older than 50 years and monitored annually thereafter.

B. Haemoglobin concentration should also be monitored, as androgen replacement can cause polycythaemia.

Testosterone replacement inhibits spermatogenesis;

20.23 Options for androgen replacement therapy

Route of administration	Preparation	Dose	Frequency	Comments
Intramuscular	Testosterone undecanoate	1000 mg	Every 3 months	Smoother profile than testosterone enantate, with less frequent injections
	Testosterone enanthate	50–250 mg	Every 3–4 weeks	Produces peaks and troughs of testosterone levels that are outside the physiological range and may be symptomatic
Transdermal	Testosterone gel	50–100 mg	Daily	Stable testosterone levels; transfer of gel can occur following skin-to-skin contact with anotl